

Assignment No.: 11

Problem Statement: Build scaling plans in AWS that balance the load on different EC2 instances.

Step1: Login into AWS account and create a template. Select Ubuntu in quick start.

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - required
keshvi123

Template version description
keshvi123

Auto Scaling guidance [info](#)
☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► Template tags
► Source template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ **Application and OS Images (Amazon Machine Image) - required** [info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below.

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64... [read more](#)
ami-084568b433236484

Virtual server type (instance type)
t2.micro

Firewall (security group)
sg-955

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage for 12 months where t2.micro isn't available when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 50 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet.

Cancel Create launch template

Step2: Select key pair and select existing security group in Firewall (security groups).

Create launch template

▼ **Instance type** [info](#) [Get advice](#) [Advanced](#)

Instance type
t2.micro
Family: t2, 1 vCPU, 1 GiB Memory, Current generation, Intel, On-Demand Windows base pricing: 0.0162 USD per Hour, Free tier eligible, All generations, Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ **Key pair (login)** [info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name
key1 [Create new key pair](#)

▼ **Network settings** [info](#)

Subnet [info](#)
Don't include in launch template [Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) [info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Select existing security group [Create security group](#)

Security groups [info](#)
Select security groups
sg-955 sg-0646f0fba011ca7 [X](#)
sg-1c9c02ac04880412805 [Compare security group rules](#)

► Advanced network configuration

▼ **Storage (volumes)** [info](#) [Hide details](#)

EBS Volumes

Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64... [read more](#)
ami-084568b433236484

Virtual server type (instance type)
t2.micro

Firewall (security group)
sg-955

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage for 12 months where t2.micro isn't available when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 50 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet.

Cancel Create launch template

Step3: After Selecting security group go to the user data and write given cmd. Click launch instance.

Create launch template

Metadata accessible [Info](#)

Enabled

Metadata IPv6 endpoint [Info](#)

Don't include in launch template

Metadata version [Info](#)

V2 only (token required)

For V2 requests, you must include a session token in all instance metadata requests. Applications or agents that use V1 for instance metadata access will break.

Metadata response hop limit [Info](#)

2

Allow tags in metadata [Info](#)

Don't include in launch template

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose File](#)

```
#!/bin/bash
apt-get update
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -sL https://deb.nodesource.com/setup_16.x | sudo -E bash -
apt-get install -y nodejs
git clone http://github.com/sudip7407/Report.git
cd Report
npm install
node index.js
```

☐ User data has already been base64 encoded

Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64... [read more](#)
ami-084568b4383264d4

Virtual server type (instance type)
t2.micro

Firewall (security group)
guir03

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage for t2.micro where t2.micro isn't available when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet.

[Cancel](#) [Create launch template](#)

Step4:Create auto scaling group, enter name for auto scaling group, select template, click on next.

Create Auto Scaling group

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name

Enter a name to identify the group.

kschri123

Must be unique to this account in the current region and no more than 255 characters.

Launch template [Info](#)

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

kschri123

[Create a launch template](#)

Version

Default (1)

[Create a launch template version](#)

Description

kschri123

AMI ID

ami-084568b4383264d4

Key pair name

key1

Launch template

kschri123

lt-0fa22986c58c10566

Instance type

t2.micro

Security groups

-

Request Spot Instances

No

Security group IDs

sg-0b644f0faad113a27

Additional details

Storage (volumes)

-

Date created

Mon Apr 21 2025 19:31:09 (GMT+05:30 (India Standard Time))

[Cancel](#) [Next](#)

Step5:Now select all availability zones and subnets then click on next.

Create Auto Scaling group

adding instance types.

Launch template

kschri123

lt-0fa22986c58c10566

Version

Default

Description

kschri123

Instance type

t2.micro

Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-024c1f4b66412d05

172.31.0.0/16

[Create a VPC](#)

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

us-east-1a | subnet-09b02e147dbab1fb X
172.31.0.0/20

us-east-1b | subnet-0622a5812e920a15 X
172.31.16.0/20

us-east-1c | subnet-0a081f52af5508660 X
172.31.32.0/20

us-east-1d | subnet-0fa4fbd8f9a4c527 X
172.31.0.0/20

us-east-1e | subnet-0783e5034d8816e42 X
172.31.0.0/20

us-east-1f | subnet-05a88be19324f47e X
172.31.0.0/20

[Create a subnet](#)

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**

If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**

If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to ensure balanced distribution.

Step6:In load balancing select attach to new load balancer, in attach to new load balancer select applicatio load balancer, in load balancer select internet-balancing.

The screenshot shows the 'Integrate with other services - optional' step in the AWS 'Create Auto Scaling group' wizard. The left sidebar lists steps 1 through 7, with step 6 selected. The main content area is titled 'Integrate with other services - optional' and includes a sub-section 'Load balancing'. Under 'Load balancing', there are three radio button options: 'No load balancer', 'Attach to an existing load balancer', and 'Attach to a new load balancer'. The 'Attach to a new load balancer' option is selected. Below this, the 'Attach to a new load balancer' section is expanded, showing fields for 'Load balancer type' (Application Load Balancer), 'Load balancer name' (keshri123-1), 'Load balancer scheme' (Internet-facing), and 'Network mapping' (VPC: vpc-024cc9f48b05412db05). The 'Availability Zones and subnets' section shows two subnets selected: subnet-0783e50343881b6d2 and subnet-0a0d81f924f6508660.

Step7:In add 1000 in port, select new target group name. In health check period add 200 seconds.

The screenshot shows the 'Listeners and routing' step in the AWS 'Create Auto Scaling group' wizard. The left sidebar lists steps 1 through 7, with step 7 selected. The main content area is titled 'Listeners and routing' and includes a sub-section 'Listeners and routing'. Under 'Listeners and routing', there are fields for 'Protocol' (HTTP), 'Port' (1000), and 'Default routing (forward to)' (gum123-1 | HTTP). Below this, the 'Tags - optional' section is expanded, showing an 'Add tag' button. The 'VPC Lattice integration options' section shows two radio button options: 'No VPC Lattice service' and 'Attach to VPC Lattice service'. The 'No VPC Lattice service' option is selected. Below this, the 'Application Recovery Controller (ARC) zonal shift - new' section is expanded, showing an 'Enable zonal shift' checkbox. The 'Health checks' section shows an 'EC2 health checks' checkbox, which is selected, and an 'Additional health check types - optional' section with a 'Turn on Elastic Load Balancing health checks' checkbox, which is also selected.

Step8:In desired capacity write 2. In scaling limit give min 2 and max 3.

Step 1: Choose launch template
Step 2: Choose instance launch options
Step 3 - optional: Integrate with other services
Step 4 - optional: Configure group size and scaling
Step 5 - optional: Add notifications
Step 6 - optional: Add tags
Step 7: Review

Configure group size and scaling - optional [info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [info](#)
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and MemoryGiB are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances):

Desired capacity
Specify your group size.

Scaling [info](#)
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

Equal or less than desired capacity

Max desired capacity

Equal or greater than desired capacity

Automatic scaling - optional [info](#)
Choose whether to use a target tracking policy. You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

Metric type [info](#)
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Target value

Step9:In target value as 50, instance warmup as 200 sec. Then click on create auto scaling group.

Target value

Instance warmup [info](#)
 seconds

☐ Disable scale in to create only a scale-out policy

Instance maintenance policy [info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

Mixed behavior

☒ No policy
For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.

Priority availability

☐ Launch before terminating
Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.

Control costs

☐ Terminate and launch
Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.

Flexible

☐ Custom behavior
Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.

Additional capacity settings

Capacity Reservation preference [info](#)
Select whether you want Auto Scaling to launch instances into an existing Capacity Reservation or Capacity Reservation resource group.

☒ Default
Auto Scaling uses the Capacity Reservation preference from your launch template.

Step10:Go to load balancer copy DNS name and run it in another tab.

Load balancers (1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

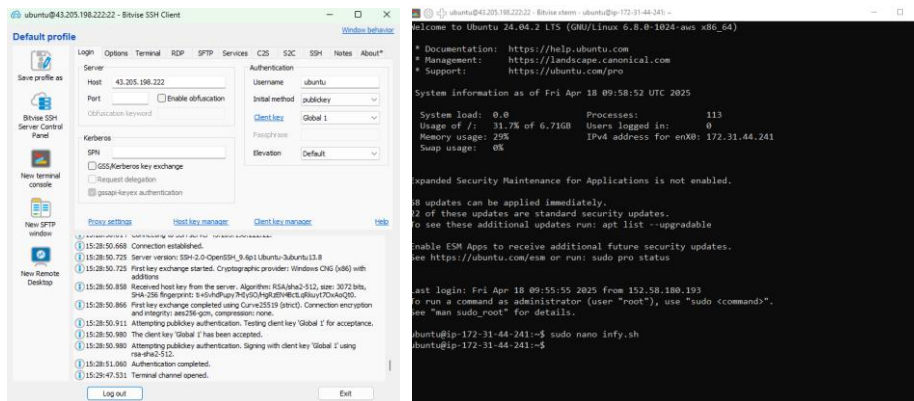
Name	State	VPC ID	Availability Zones	Type	Date created
ke3hr123-1	Provisioning	vpc-024ccf48d64412db5	6 Availability Zones	application	April 21, 2025, 19:55 (UTC+05:30)

0 load balancers selected

Select a load balancer above.



Step11: Now open bitwise and copy Public IPv4 address from any one instance and paste it in port. Open terminal and write the given command, it will start a loop.



Step12: In security we can see CPU utilization graph.

